GGG-N-202C
June 16, 1971
SUPERSEDING
Fed. Spec. GGG-N-202B
September 30, 1964

FEDERAL SPECIFICATION

NEEDLE, SAILMAKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sailmaker's needles.

1.2 Classification.

1.2.1 Size numbers. Sailmaker's needles shall be of size numbers 6, 8, 10, 11, 12, 13, 14, 15, 16, 17 and 18, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index. which includes cumulative monthly supplements as issuec, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

FSC 8315

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.4).

3.2 Material. Sailmaker's needles furnished under this specification shall be forged of cast steel to produce needles in accordance with the requirements specified herein.

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TABLE I. Dimensions and tolerances*

Needle size number	Over-all length A ± 3/16	B±5/16	C	D±1/8	E±0.005	F±0.011	G±0.007	H±0.0312 1/
	Inches	Inches	Inch	Inches	Inch	Inch	Inch	Inch
6	5-1/8	3-1/16	0.201 ± 0.016	1-1/2	0.189	0.204	0.138	0.417
8	4-9/16	2-5/8	.169 ± .014	1-3/8	.156	.164	.117	.367
10	3-3/4	2	.138 ± .012	1	.127	.134	.097	.344
11	3-7/16	1-13/16	.122 ± .012	7/8	.115	.112	.085	.287
12	3-3/16	1-13/16	.108 ± .010	7/8	.102	.101	.077	.229
13	3	1-5/8	.096 ± .010	3/4	.090	.088	.067	.205
14	2-3/4	1-1/2	.085 ± .008	3/4	.078	.080	.060	.180
15	2-5/8	1-1/4	.075 ± .008	5/8	.070	.068	.050	.173
16	2-7/16	1-1/4	.068 ± .006	5/8	.063	.064	.047	.166
17	2-5/16	1-1/8	.060 ± .006	9/16	.054	.057	.039	.149
18	2-3/16	1-1/8	.052 ± .004	1/2	.046	.054	.035	.133

1/ Inside length of needle eye.

* See table III and figures 1 and 2.

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3.3 Design. Each needle shall be ground to a fine point. The edges of the triangular section (see figure 1, section X-X) shall be ground and rounded in such a manner that material fibers shall not be cut when pierced by the needle in the normal sewing operation. The eye (see figure 1, section Z-Z) shall be rounded and so designed that the thread will not be cut when used under ordinary service conditions and the opening of the eye shall be centered between its lateral surfaces. Needles shall be heat treated to meet the deflection requirements of table III without permanent set.

3.4 Dimensions. Needles shall be similar to figure 1 and shall conform to the dimensions shown in table I. In addition the eye opening shall accommodate the applicable size thread.

3.5 Finish. Needle surfaces shall be smooth and polished. The eye of the needle shall be smooth. The section of the eye adjacent to the heel shall be polished.

3.6 Workmanship. The finished needle shall be free of burrs, slivers, cracks, nicks, pit holes and scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, it shall be examined for the defects specified in table II, dimensions specified, and tested as specified in 4.4.1. Any visual defect, any dimension not meeting specified requirements, or failure to meet the test requirements, shall be cause for rejection of the preproduction sample.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.2 End item inspection. The lot shall consist of all the needles of one size offered for inspection at one time. The sample unit shall be one needle.

4.3.2.1 Visual examination. Examination shall be made of the end item for defects listed in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Surface (including periphery of the eye) is not smooth	X	
	Evidence of imbedded foreign matter		X
Construction and workmanship	Does not have fine sharp point	X	
	Edges of triangular not ground and rounded		X
	Eye not centered between lateral edges	X	
	Eye is closed	X	
	Eye is partially closed		X
	Sharp edge at eye (inside or outside)	X	
	Burrs or slivers not removed	X	
	Needle is cracked, or broken	X	
	Evidence of pitholes, nicks, or scratches		X

4.3.2.2 Dimensional inspection. Inspection shall be made of needles for compliance with dimensions specified. Any deviation therefrom shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units.

4.3.2.3 Testing of end item. Needles shall be tested a minimum of 3 times once on each plane as specified in 4.4.1. Any deviation therefrom shall constitute a defect. The inspection level shall be S-4 with an AQL of 2.5 expressed in terms of defects per hundred units.

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4.4 Tests.

4.4.1 Deflection tests. The point end of the needle shall be placed between two pieces of 3/16 inch diameter drill rod, held rigidly by a vise, as shown in figure 2, or by a vise with V Block, or by other means. The distance "s" between the drill rods shall be as shown in table III. The tip of the needle shall extend 3/32 inch beyond the point of contact with the drill rod. When the needles are held by other than drill rods, they shall be gripped 7/32 maximum from the point of the needle. A force shall be applied to the eye end of the needle, as close as possible to the heel, in a direction that will cause the needle to bend while the point remains stationary. A relatively sharp edged tool such as a screwdriver or chisel shall be used in applying this force. The heel of each needle shall be deflected a linear distance "d" as specified in table III. This test shall be performed for each of the three planes represented by the triangular section of each needle. Any permanent set of needle shall constitute failure of this test.

TABLE III. Distance and reflection

Needle number	Distance(s)	Distance(d)
	Inch	Inch
6	7/16	9/16
8	7/16	9/16
10	7/16	9/16
11	3/8	5/8
12	3/8	5/8
13	3/8	9/16
14	3/8	9/16
15	3/8	9/16
16	3/8	9/16
17	5/16	9/16
18	5/16	9/16

4.5 Examination of preparation for delivery. Examination of preparation for delivery shall be made in accordance with PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing for level A, B or C shall be in accordance with PPP-P-40 (see 6.2). Size 14 needles shall be packaged in quantities of 25 per unit package.

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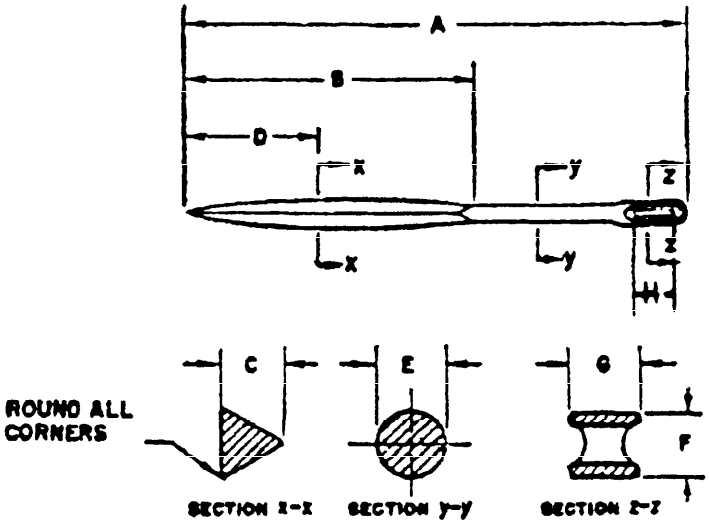


Figure 1. Sailmaker's needles.

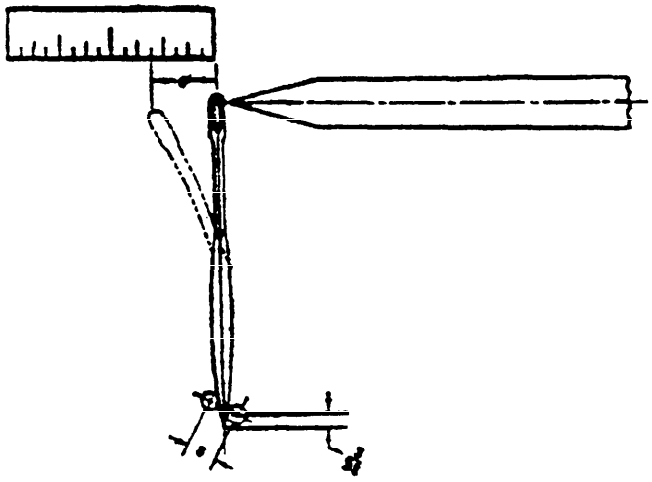


Figure 2. Deflection test.

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5.2 Marking.

5.2.1 Civilian agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The needles covered in this specification are intended to be used by sailmakers for sewing and repairing canvas.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Sizes of sailmaker's needles required (see 1.2).
(c) When a first article is required (see 3.1, 4.2 and 6.4).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Needle size numbers. The needle size numbers referenced in this specification are the applicable "English Wire Gage" numbers.

6.4 First Article. When a first article is required it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample of the specified number size needle. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

GSA-FSS
COM-NBS
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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Needle, Sailmaker's [illegible]

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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